

Hariharan T S

Software Engineer | Full Stack Development · Python/Django/FastAPI · React | AI/ML
hari04022005@gmail.com | +91 6382939578 | Madurai, India | LinkedIn | GitHub | Portfolio

Professional Summary

Software Engineer with hands-on experience designing and building complete applications end-to-end, from backend architecture and APIs to frontend/mobile interfaces and AI/ML integration. Skilled in Python backend development using Django, Django REST Framework, and FastAPI; REST API design; SQL and database integration; and modern frontend/mobile development. Experienced with Machine Learning, Deep Learning, TensorFlow, Scikit-learn, NLP, OCR, LLM integration, Prompt Engineering, RAG pipelines, vector embeddings, FAISS, and ChromaDB. Strong problem-solver with 500+ LeetCode problems solved and a solid foundation in data structures, algorithms, OOP, DBMS, software engineering, and system design.

Technical Skills

Programming Languages: Python, Java, SQL

Backend Development: Django, Django REST Framework, FastAPI, REST APIs, CRUD, Async Programming, API Validation

Frontend & Mobile: React.js, HTML5, CSS3, Responsive Design

Databases: PostgreSQL, MySQL, SQLite, Firebase

AI & Machine Learning: Machine Learning, Deep Learning, TensorFlow, Scikit-learn, NLP, OCR, CNN, Model Evaluation, Feature Engineering

Generative AI / LLM & RAG: Generative AI, LLM Integration, Prompt Engineering, RAG Pipelines, LangChain (Familiar), OpenAI API (Familiar), Vector Embeddings, FAISS, ChromaDB, Semantic Search

Cloud & DevOps: Git, GitHub, GitHub Actions (CI/CD), Firebase, Vercel, AWS Fundamentals (EC2, S3, Lambda)

Software Engineering: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, SDLC, System Design, API Design, Problem Solving

Projects

AI-Powered Personal Finance Management System 🔒

Python, FastAPI, Flutter, SQLite, Firebase, TensorFlow, Scikit-learn, OCR, NLP, RAG, Vector Embeddings

- Architected an end-to-end AI-powered financial assistant with a FastAPI backend and Flutter cross-platform frontend using a modular architecture.
- Developed 12+ modular asynchronous REST APIs with CRUD operations, request validation, and JWT-based authentication for financial data management.
- Built OCR and NLP pipelines for receipt extraction and SMS-based expense parsing, feeding TensorFlow/Scikit-learn models achieving approximately 91% accuracy on 5,000+ transactions.
- Integrated an LLM through a RAG pipeline using vector embeddings for semantic retrieval and grounded financial insights through prompt engineering.
- Implemented a responsive Flutter UI with real-time expense analytics and offline-first SQLite/Firebase synchronization supporting 10,000+ records.
- Maintained version-controlled releases through GitHub-based CI/CD practices.

Stealth Track – AI-Powered Habit & Productivity Platform 🔒

Python, Django, Django REST Framework, React Native, PostgreSQL, JWT, Scikit-learn

- Engineered a full-stack productivity application with a Django REST Framework backend and React Native mobile frontend.
- Designed REST APIs for habit/task CRUD, analytics, and user management with PostgreSQL, JWT authentication, and authorization.
- Integrated Scikit-learn for AI-powered habit and task categorization and built analytics endpoints for productivity insights.
- Implemented recurring task scheduling, automated notifications, streak tracking, and gamification features.
- Structured the backend using reusable serializers and service-layer logic following Django best practices.

Interactive Data Explorer – One Piece Encyclopedia

React.js, JavaScript, Context API, Canvas API, JSON, Lazy Loading, Vercel

- Built a complete React web application with dynamic routing and a normalized data model representing 200+ entities and 30+ story arcs.

- Implemented global state management with Context API across 15+ routes and interactive Canvas API visualizations for relationship mapping.
- Optimized frontend performance by approximately 40% using lazy loading and memoization; deployed on Vercel with GitHub CI integration.

Traffic Sign Detection – CNN Classifier

Python, TensorFlow, CNN, OpenCV, Scikit-learn, GTSRB Dataset

- Trained a CNN on 50,000+ images across 43 classes, achieving 96.4% test accuracy.
- Applied data augmentation and class-weighted loss functions to handle class imbalance, reducing misclassification by 22%.
- Evaluated performance using precision, recall, F1-score, and confusion-matrix analysis.

Education

Sethu Institute of Technology, Madurai

B.E. Computer Science & Engineering (AI & ML)

- CGPA: 8.79 / 10 | Honors: Full Stack Development
- Relevant Coursework: Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, DBMS, Computer Vision, Software Engineering

Thiagarajar Model Higher Secondary School, Madurai

Class XII: 84%

Thiagarajar Model Higher Secondary School, Madurai

Class X: 83.4%

Certifications

- AI Master Class (30-day Hands-on Program) – Novitech Pvt Ltd
- Infosys Virtual Internship – December 2025 – May 2026
- AI Primer Certification – Infosys Springboard
- Project-Based Robotics Training (2 programs) – Zetspire Technologies Pvt Ltd